

# Agentic AI for Biomedical Data Science

## CCB Workshop

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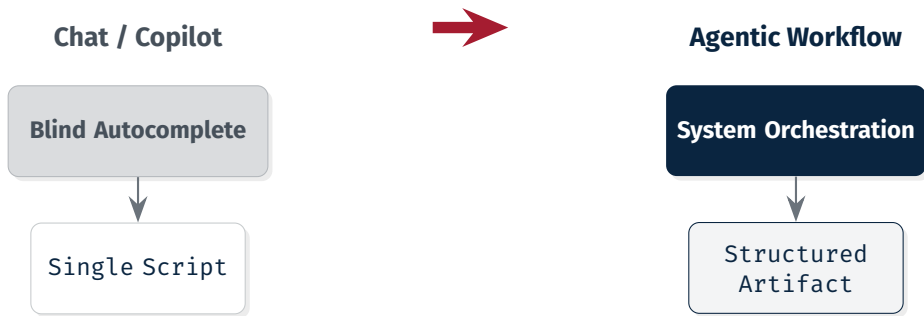
**HARVARD**  
MEDICAL SCHOOL

**BLAVATNIK INSTITUTE**  
BIOMEDICAL INFORMATICS

September 24, 2026

# The Paradigm Shift in AI

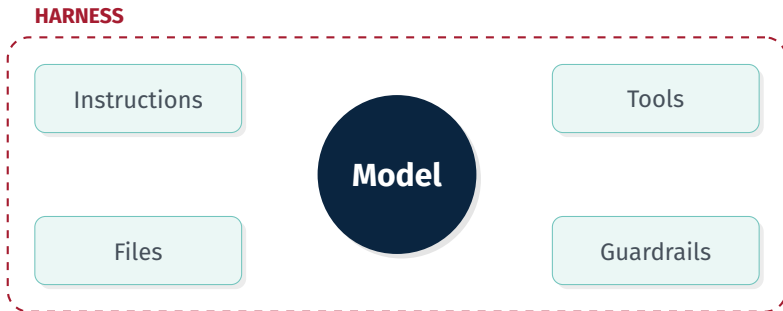
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*The harness is the program you write around the model.*

# What You Actually Control

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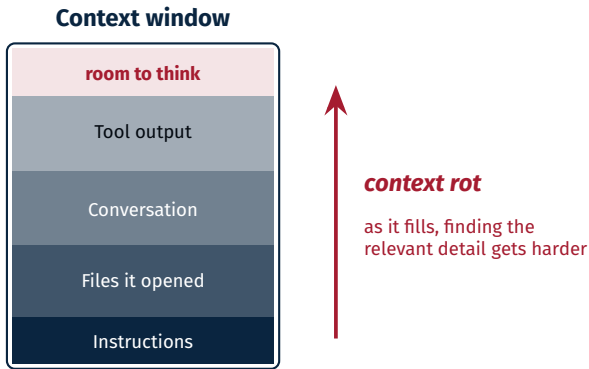


*You don't control the model. You control the text.*

↪ Anthropic: Building Effective Agents · HumanLayer: Harness Engineering

# Context: The Finite Resource

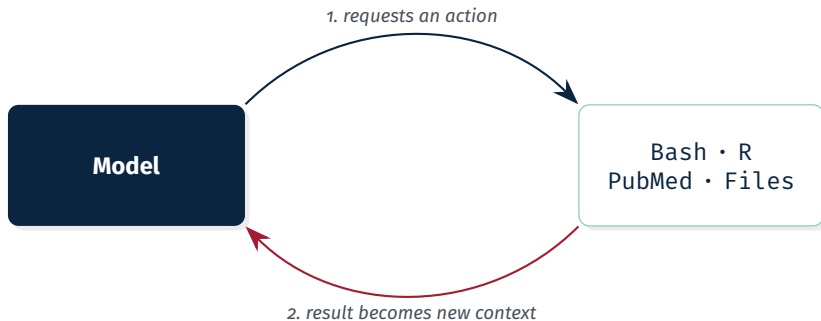
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*A bigger window is a bigger haystack — not a better search.*

# Tool Calling: How an Agent Acts

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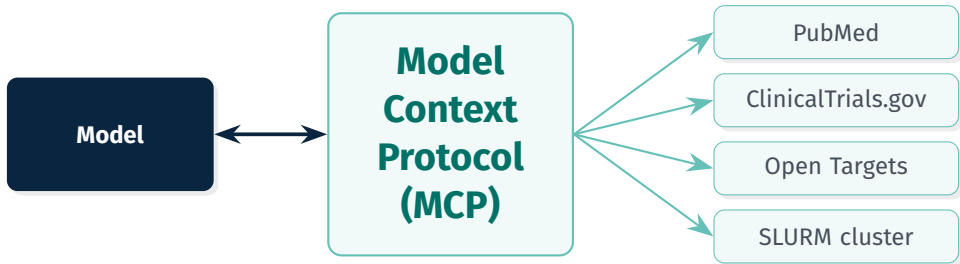


*Talks about DESeq2 → runs DESeq2.*

↪ Anthropic: Writing Effective Tools for Agents

# One Standard, Many Tools

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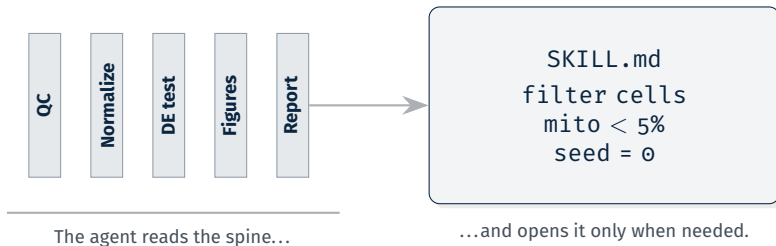


*Plug in once — no custom wiring for every tool.*

↪ [modelcontextprotocol.io](https://modelcontextprotocol.io) · Claude Code: MCP

# Skills: Know-How on a Shelf

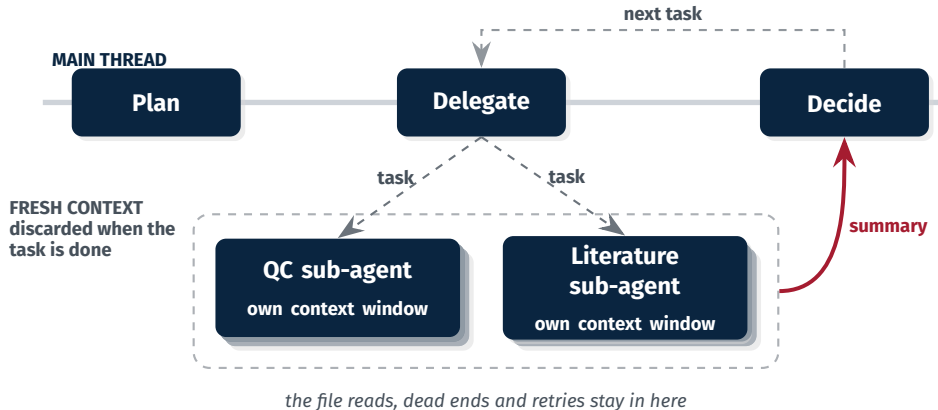
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*Progressive disclosure: your lab's protocols, written down once.*

↪ Anthropic: Agent Skills · Claude Code: Skills

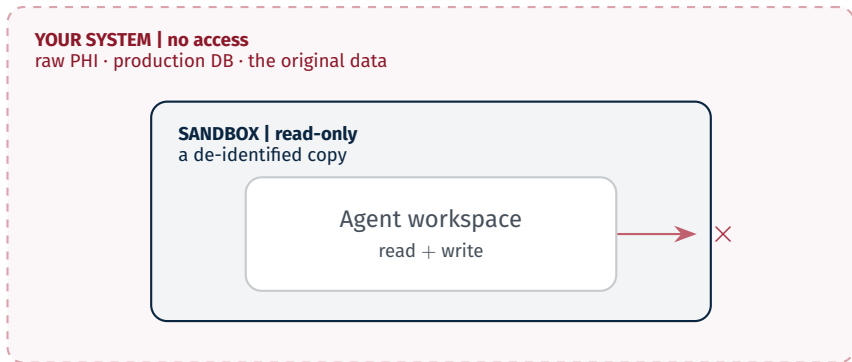
# Sub-Agents: Delegate the Task



*Only the summary comes back. The clutter never reaches the main thread.*

# If PHI Is Involved

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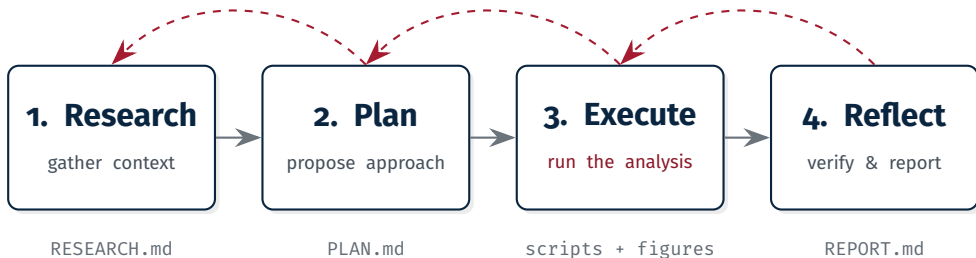
*Least privilege: the agent gets a copy, not the original.*

↪ Claude Code: Sandboxing · Apptainer · Reproducibility standards for ML in the life sciences

# The Agentic AI Framework

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Human review at every step: send it back one phase



# Setup: Configuration of the Agent

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## General Setup

```
project/  
  CLAUDE.md  
  .claude/skills/  
    <skill>/SKILL.md
```

*generated once*

## Project-Specific Setup

```
project/analyses/  
  <analysis-1>/  
    RESEARCH.md  
    PLAN.md  
    REPORT.md  
  <analysis-2>/ ...
```

*agent generated*

# CLAUDE.md: General Setup

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**CLAUDE.md** — *instructions for standards, tooling and rules.*  
project/CLAUDE.md

## **## Tooling**

QC in Scanpy · stats in R (DESeq2)

## **## Standards**

Lab figure style · seed = 0 everywhere

## **## Rules**

Never edit data/raw/

De-identify before any external tool

*Notice what's not here: no biology, no experiment specifics — that belongs in RESEARCH.md.*

# SKILL.md: A Protocol, Written Down Once

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**SKILL.md** — *every skill opens with a name and a description.*

project/.claude/skills/scrna-qc/SKILL.md

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**name:** scrna-qc

**description:** QC filtering for scRNA-seq.

Use when preparing a new dataset.

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1. Drop cells with mito > 5%
2. Drop genes seen in < 3 cells

*You never name the skill — the description is how the agent finds it.*

## Phase 1: Research (Context First)

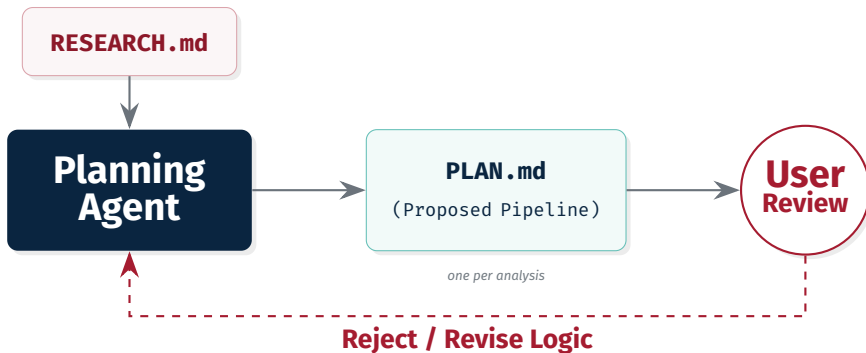
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↪ Biomni: a general-purpose biomedical AI agent

## Phase 2: Plan

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# Model Choice: Expensive to Plan, Cheap to Execute

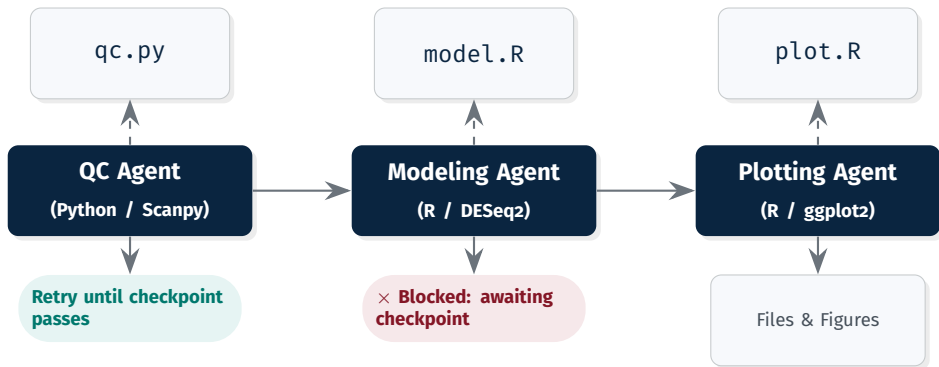
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*Planning is the complex part — worth investing in. Once the plan is clearly specified, a cheaper model can carry it out.*

## Phase 3: Execute (Sub-Agents & Blocking)

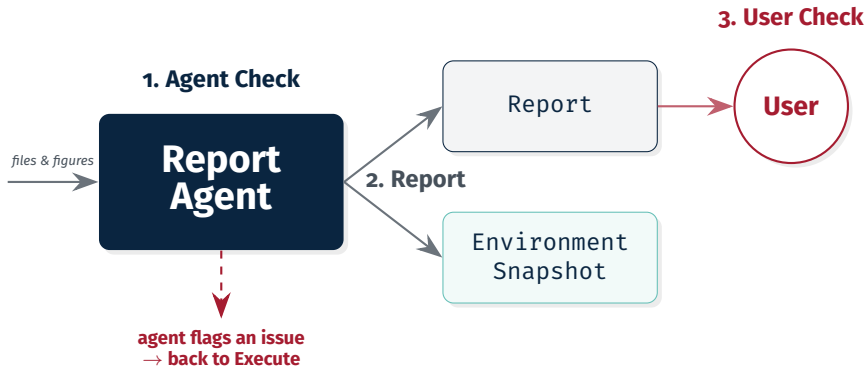
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*Runs the approved PLAN.md. Halts automatically if a checkpoint — set in CLAUDE.md or a SKILL.md — is not met.*

## Phase 4: Automated & User Checks

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*The agent checks its own work first. You still make the final call before it's done.*

# What Can Go Wrong

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**Completion  $\neq$  Correctness**

**Plan can hide  
real complexity**

**Can delete  
real files**

**Could be a  
hallucination**

**Don't trust  
blindly**

↪ HumanLayer: Harness Engineering · Kapoor & Narayanan: Leakage & the reproducibility crisis

## Best Practices

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- ▶ Keep **CLAUDE.md** short and accurate.
- ▶ Review the plan before anything executes — make sure you and the agent are aligned on the analysis goals.
- ▶ Be very careful with patient or sensitive data — give the agent only the access it needs.
- ▶ An agent can help with git/version control — but it needs guardrails (no force-push, no exposed keys).

## Sensitive Data: Best Practices

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- ✓ Test and demo with de-identified or synthetic data
- ✓ Keep real patient data on your institution's approved, secure systems
- ✓ Ask your data-security office when you're not sure

## Sensitive Data: What Not To Do

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- ✗ Paste patient identifiers into a public AI tool
- ✗ Let an agent send data anywhere without checking where it goes
- ✗ Assume an agent changes the rules — every rule for sensitive data still applies

*Treat the agent like any new team member: need-to-know access only.*

# Data Levels: Match Your Data to the Right Tool

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**1 Public AI tools permitted** — public, non-sensitive information only

**2 HMS-approved meeting assistants only** — e.g. Zoom Companion, Copilot in meetings

**3 Harvard-licensed tools** (Sandbox, Claude, ChatGPT Edu, Copilot) — contractual safeguards in place

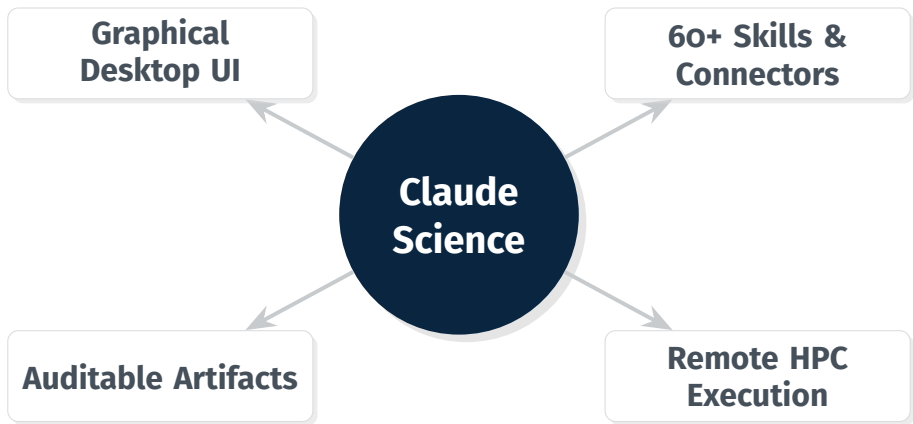
**4 HMS Azure AI / Longwood Cluster** — approval required: [hms-it-ai@hms.harvard.edu](mailto:hms-it-ai@hms.harvard.edu)

**5 No approved AI tool** — highest-risk regulated data is off-limits

*Need agentic AI on Level 4–5 data? Ask CCB about running a local, self-hosted model instead.*

# Claude Science Workbench

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↔ Claude Science (AI workbench, 2026) · Claude for Life Sciences (2025)

## Summary

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- ▶ A **structured agentic workflow** beats clever prompt engineering.
- ▶ **Separate** Research and Plan from Execution.
- ▶ Use **CLAUDE.md** and **SKILL.md** for general configuration.
- ▶ Use **RESEARCH.md**, **PLAN.md** and **REPORT.md** for their phases.
- ▶ Keep a **human in the loop** at every stage of the workflow.

# Go Deeper: Concepts & Tools

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## TALKS & WRITEUPS BEHIND THIS SESSION

- ↪ “Slop Is a Skill Issue” (talk)
- ↪ HumanLayer: Harness Engineering
- ↪ HumanLayer: Advanced Context Engineering
- ↪ “Learning While You Sleep” (talk)

## CORE CONCEPTS

- ↪ Anthropic: Building Effective Agents
- ↪ Anthropic: Effective Context Engineering
- ↪ Anthropic: Writing Effective Tools for Agents
- ↪ Anthropic: Multi-Agent Research System
- ↪ Anthropic: Equipping Agents with Skills

## CLAUDE CODE DOCUMENTATION

- ↪ Claude Code: Overview
- ↪ Claude Code: Best Practices
- ↪ Claude Code: The Context Window
- ↪ Claude Code: Subagents
- ↪ Claude Code: Skills
- ↪ Claude Code: Hooks
- ↪ Claude Code: MCP
- ↪ Anthropic: Agent Skills
- ↪ [modelcontextprotocol.io](https://modelcontextprotocol.io)

# Go Deeper: Agentic AI in Biomedicine

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## AGENTS FOR BIOMEDICAL RESEARCH

- ↪ Biomni: a general-purpose biomedical AI agent
- ↪ The Virtual Lab (Nature, 2025)
- ↪ CellAgent: single-cell multi-agent framework
- ↪ Qiu & Topol: Agentic systems in medicine
- ↪ BixBench: benchmark for computational biology

## FROM OUR OWN DEPARTMENT (ZITNIK LAB)

- ↪ ToolUniverse (Zitnik Lab, HMS DBMI)
- ↪ ToolUniverse (GitHub)
- ↪ TxAgent (Zitnik Lab)

## REPRODUCIBILITY

- ↪ Reproducibility standards for ML in the life sciences
- ↪ Kapoor & Narayanan: Leakage & the reproducibility crisis

## ENVIRONMENTS & CONTAINERS

- ↪ conda · Bioconda
- ↪ renv
- ↪ Docker · Apptainer

## PRODUCTS

- ↪ Claude Science (AI workbench, 2026)
- ↪ Claude for Life Sciences (2025)